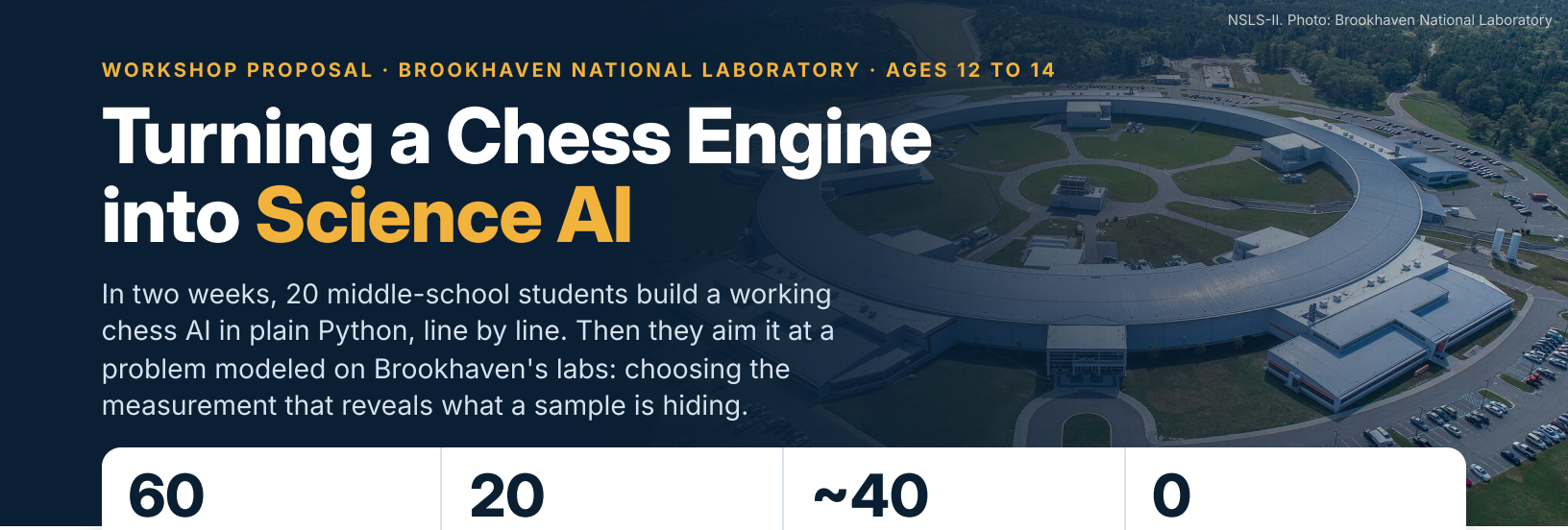

NSLS-II. Photo: Brookhaven National Laboratory

WORKSHOP PROPOSAL · BROOKHAVEN NATIONAL LABORATORY · AGES 12 TO 14

Turning a Chess Engine into Science AI

In two weeks, 20 middle-school students build a working chess AI in plain Python, line by line. Then they aim it at a problem modeled on Brookhaven's labs: choosing the measurement that reveals what a sample is hiding.

60	20	~40	0
class hours over 10 days, four 90-minute periods a day	students in 10 pairs, about five pairs per adult	engine functions written by the students themselves	student online accounts; everything runs offline on class laptops

Chess is where AI learned to think ahead. Science is where it is going next.

Every game-playing AI runs the same loop: describe the situation, list the possible actions, look ahead, score the results, and check the answer. That loop is also how an AI agent decides which measurement to take next at a beamline.

Students build **Remus**, a complete chess engine with no hidden libraries. They write the rules, verify them against a trusted reference, and teach the engine to search and judge positions. On Day 9 each pair adds one enhancement and measures its effect in unattended matches; the finished class engine rates about 1520 Elo against Stockfish.

The goal is not chess strength. Students practice three habits of scientific computing: they **build a model, test it against a reference, and measure the results**.

THE LINEAGE STUDENTS WILL TRACE

1978 — **Sargon** chess program publishes its full source as a book

1987 — **Turbo GameWorks Chess**: the readable engine Remus is modeled on

1997 — **Deep Blue** defeats world champion Garry Kasparov

2017 — **AlphaZero** teaches itself chess by self-play and search

2021 — **NSLS-II** scientists teach an AI agent to "play" experiments

2024 — **Nobel Prize in Chemistry** honors AlphaFold protein structure AI

2025 — **DOE Genesis Mission** launches AI for science across the national labs

SAME LOOP, TWO WORLDS

	Represent	Generate	Look ahead	Evaluate	Verify
IN REMUS	An 8×8 board and the full game state	Every legal move, including en passant	Minimax and alpha-beta search	Score material and position	Perft counts match a trusted reference
IN SCIENCE AI	A sample, a model, or an instrument state	Candidate measurements or hypotheses	Predict outcomes before spending beam time	Score how well each hypothesis explains the data	Compare predictions with measured data

Built for a national laboratory classroom

- **Plain, readable Python.** Standard library only, so nothing is a black box.
- **Real software practice.** Tests written before code, daily checkpoints, code review in pairs.
- **Hands-on first.** Paper-and-card warm-ups before each hard idea.
- **Rooted in Brookhaven science.** Scientist visits connect search and evaluation to AI at NSLS-II.

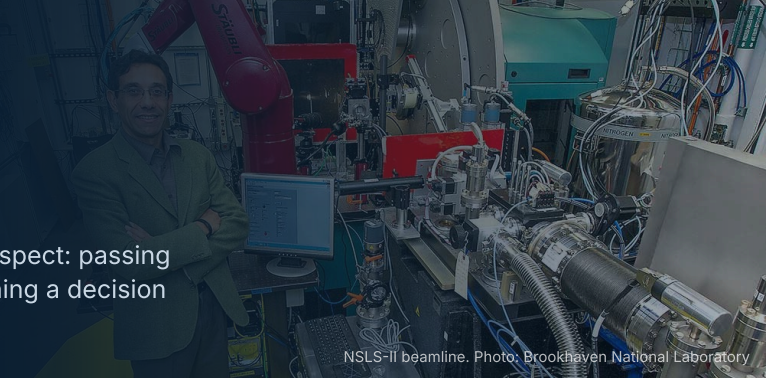
Safe and low-friction for minors

- **No accounts, no cloud.** Preconfigured Windows 11 laptops work offline.
- **Code goes home on USB.** Each student keeps their engine and every day's code.
- **Open and reusable.** The course and engine are BSD-licensed for other sites to adopt.

THE CASE FOR SUPPORT

Measured outcomes, not promises.

Every skill the course teaches leaves evidence a reviewer can inspect: passing tests, matching counts, credited functions, and a student explaining a decision their own code made.



NSLS-II beamline. Photo: Brookhaven National Laboratory

What students will be able to do AND HOW WE WILL KNOW

OUTCOME	EVIDENCE A REVIEWER CAN CHECK	WHEN
Model a system as data	Board and game state code; ten moves made and undone return the exact starting state	Day 2
Write code to a specification	Each pair's functions pass teacher-written tests; a credits file shows who wrote what	Days 1-10
Verify against a reference	Move-tree counts of 20, 400, and 8,902 positions match an independent library	Day 6
Reason about search	Fewer nodes searched after alpha-beta and move ordering, same move chosen	Day 8
Measure a change	Rating gain of each pair's enhancement over 20 games against the base engine	Day 9
Explain a result in plain words	Each pair explains one engine move (score and line of play) and one scan choice	Day 10
Find a hidden peak	Each pair's measurement AI locates a hidden hot spot in fewer readings than random guessing, over 100 fixed trials	Day 10
Name the crystals in a pattern	On 200 X-ray patterns with known answers, each pair's tree search names the crystal phases with fewer wrong calls than a greedy guess, and flags the patterns it cannot decide	Day 10

PRE AND POST
The same 20-minute concept check on Day 1 and Day 10

ARTIFACTS
Tests passed and perft matches per pair, read on each laptop

RUBRIC
Correctness, evidence, explanation, and teamwork

INTEREST SURVEY
Interest in computing and science careers, before and after

A first step on the Brookhaven pipeline STEM WORKFORCE

AGES 12 TO 14
Turning a Chess Engine into Science AI

MIDDLE AND HIGH SCHOOL
Science Bowl and Summer Science Explorations

HIGH SCHOOL
High School Research Program

GRADUATES AND FRESHMEN
WDTS Big Data Science and Applications School

UNDERGRADUATE
DOE laboratory internships

Mission alignment

- **DOE Genesis Mission** (launched November 2025) aims to double the productivity and impact of U.S. science within a decade through AI. This workshop shows young students what AI for science is made of.
- **DOE Office of Science** user facilities such as NSLS-II already run AI agents that choose the next measurement.

What the pilot needs

- **Space and time:** one classroom for 10 days, 8:30 to 4:30.
- **Equipment:** 10 preconfigured laptops and 20 USB drives.
- **People:** one teacher, one assistant, and two or three short scientist visits.
- **Curriculum:** complete and open on GitHub (dbiersach/Remus).

The decision

1. **Approve a pilot** of the workshop for 20 students, designed to grow to 30.
2. **Arrange scientist visits** at NSLS-II, and share one measured two-phase powder pattern from the PDF or XPD beamline as the final test for Day 10's phase hunt, a phase identification search modeled on published synchrotron work.
3. **Support the evaluation** with Brookhaven education staff: pre and post checks, rubric, and survey.
4. **Judge it by evidence:** student code, tests, explanations, and growth, not by chess strength.